

Customer Data Enrichment and Behavioral Segmentation of Purchasing Data

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Customer Data Enrichment and Behavioral Segmentation of Purchasing Data

The Purchasing_data workbook contains 4,362 customer survey records and 89 variables collected from September 15 to September 24, 2021. It captures credit-card ownership, balances, repayment, share of purchases, Buy Now, Pay Later (BNPL) use, and demographic and financial characteristics. Its age, self-reported design, and questionnaire routing, however, limit its usefulness for current targeting. This report proposes a defensible enrichment strategy and applies latent class analysis (LCA) to identify actionable customer groups. The analysis separates class-forming behaviors from demographic variables used only for external profiling, reducing the risk that the segments merely reproduce age or income categories.

Part A: Data Enrichment Plan

Missing and Outdated Information

The principal weakness is temporal relevance. Responses were collected in 2021, while the income question refers to 2020; payment technology, prices, household finances, and channel preferences have since changed. The file provides snapshots over different recall periods but no longitudinal history, transaction recency, purchase frequency, monetary value, merchant category, promotional response, app activity, or channel engagement. It also lacks psychographic measures such as convenience orientation, price sensitivity, risk tolerance, attitudes toward debt and BNPL, financial confidence, product involvement, brand loyalty, switching intention, privacy preference, and financial goals. Household composition, life stage, employment change, current income, preferred communication channel, and consent status should also be refreshed. Structural blanks created by survey routing must remain distinguishable from true nonresponse. Together, these additions would support behavioral, needs-based, value, and engagement segmentation rather than relying on dated demographics alone.

Proposed Data Sources

The primary source should be a consent-based customer survey delivered through the loyalty application and email, with a telephone option for less digitally engaged customers. A quarterly pulse would update channel use, life changes, product needs, promotional responsiveness, and satisfaction; an annual core would measure psychographics, loyalty, financial confidence, and preferences. With meaningful consent, responses could be linked to the existing customer identifier so stated preferences can be compared with account history.

The secondary source should be the Bank of Canada's 2024 Methods-of-Payment Survey. Its representative questionnaire and three-day diary provide benchmarks for cash, cards, online payments, and emerging alternatives. Mobile payments approached 5% of point-of-sale transactions in 2024 while other alternatives continued to grow (Felt et al., 2025). Age, income, and regional benchmarks would distinguish company-specific movement from wider market trends.

Collection Method and Data Quality

Questions should use plain language, one construct, balanced scales, explicit time frames, and 'not applicable' options. Cognitive interviews and a pilot should check interpretation, completion time, missingness, reliability, routing, mobile usability, and duplicates. A version-controlled data dictionary would preserve consistent coding. AAPOR recommends aligning mode, sampling frame, questionnaire, and nonresponse procedures with clear objectives (American Association for Public Opinion Research [AAPOR], n.d.). Contact information should be separated, access restricted, and retention limited. PIPEDA also requires customers to understand the purpose and consequences of collection and to receive meaningful choice for optional analytics uses (Office of the Privacy Commissioner of Canada, n.d.).

Sampling Plan

A stratified random sample should be drawn from the current customer registry using recent card-use status, age, region, tenure, and customer-value tier. Small strategic strata may be oversampled and weighted back to population totals. Invitations should use the preferred channel, two standardized reminders, an equal incentive, and a mixed-mode alternative to reduce coverage bias. Validity would follow from probability selection, pretesting, and comparison with account benchmarks. Reliability would rely on an unchanged core instrument, repeated measures, documented coding, and scale checks. Generalizability would require population coverage, design and nonresponse weights, and transparent response-rate reporting. A rotating panel could measure change while refreshing participants to limit panel conditioning.

Part B: Customer Segmentation Analysis

Variables and Preprocessing

Five behavioral variables formed the classes: number of cards (Q02), usual monthly balance (Q03), frequency of paying in full (Q04), card share of purchases (Q05), and BNPL awareness and use (Q10). They represent relationship breadth, balance intensity, repayment, share of wallet, and adoption of newer credit. Demographics were reserved for profiling. Cleaning trimmed spaces, standardized PayPal, converted the invalid age response 'm' to missing, and found no exact duplicates. Routing explained most blanks, so none were imputed. The analysis retained 2,094 active card users (48.0%) with complete indicator responses. Categories were encoded as consecutive integers for poLCA but modeled as nominal, not continuous, measurements.

Segmentation Method and Number of Classes

LCA was selected because it estimates segments directly from multivariate categorical responses and returns class-conditional probabilities and posterior membership uncertainty (Linzer & Lewis, 2011). Models containing one through 10 classes were estimated with 50 randomized starts to reduce local-optimum risk. Bayesian information criterion (BIC) declined from 34,520.37 for one class to its minimum of 34,037.74 for four classes, then increased to 34,129.02 for five classes. The four-class solution therefore provided the strongest balance between fit and parsimony. Average maximum posterior membership was 82.6%; 15.4% of assignments fell below the 60% review threshold, so these cases should not receive rigid automated treatment.

Segment Profiles and Marketing Actions

High-Use Transactors included 637 respondents (30.4%). The model estimated a 63.4% probability of always paying the full balance and a 58.8% probability of placing at least 80% of purchases on a credit card. Externally, 71.7% had a high credit score, 54.0% had income of at least \$75,000, and only 6.3% reported difficulty paying monthly bills. Retention should emphasize premium rewards, travel or purchase protection, personalized merchant offers, and recognition benefits instead of balance-building messages.

Light or Occasional Users included 667 respondents (31.9%). They were most likely to own one card (41.4%), hold a balance below \$500 (52.7%), and place less than 20% of purchases on cards (45.4%). Only 26.7% were in the higher-income group and 21.6% reported financial strain. A low-pressure activation program should use simple category bonuses, digital-wallet onboarding, fee transparency, and reminders of existing benefits; unsolicited credit-limit expansion would be poorly aligned with this group's observed engagement.

BNPL-Enabled Growth Users comprised 334 respondents (16.0%). The probabilities of using BNPL a few or many times were 52.5% and 44.1%; 71.0% reported increased card use, 75.1% earned at least \$75,000, and 41.3% were younger than 35. Digital installment offers fit this group, but require total-cost disclosure, reminders, budget controls, and opt-out.

Balance-Carrying Revolvers comprised 456 respondents (21.8%). Rarely or never paying in full totaled 70.7%, and a balance above \$10,000 had a 25.6% probability. Only 24.8% had a high credit score, while 28.1% reported bill difficulty. Supportive treatment should prioritize balance alerts, lower-cost repayment options, hardship resources, and clear interest disclosures rather than aggressive credit cross-selling.

Validation, Limitations, and Recommendations

All seven external variables differed significantly after multiplicity adjustment, with Cramer's V from .145 to .280. K-means produced a .227 silhouette and .575 adjusted Rand index versus LCA: broad structure existed, but assignments were method-dependent. No manifest-variable pair violated local independence after Holm adjustment.

Limitations include an unknown original sampling design, self-report error, cross-sectional data, 2021 field dates, 2020 income, and restriction to active card users. LCA assigns uncertain customers to a most-probable class, while associations remain descriptive and campaign outcomes were unavailable. Segments should therefore guide testable hypotheses, not pricing or adverse credit decisions. Data should be refreshed, customers scored probabilistically, segment drift and protected-group outcomes monitored, and treatments evaluated through randomized campaign tests. These controls make the four segments a practical, responsible foundation for differentiated engagement.

References

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Table 1*Questionnaire Routing and Analytical Sample*

Sample category	n	%
Eligible active credit-card users	2,094	48.0%
No credit-card use in the past 12 months	1,631	37.4%
Zero usual monthly credit-card balance	555	12.7%
Card selected but core responses are incomplete	43	1.0%
No recent credit-card purchase use	39	0.9%
Total	4,362	100.0%

Note. Structural routing categories are mutually exclusive. The LCA used only eligible active credit-card users with complete manifest-variable responses.

Table 2*Behavioral Variables Used to Form the Latent Classes*

Variable	Construct	Marketing relevance
Q02	Number of credit cards	Relationship breadth and competitive ownership
Q03	Usual monthly balance	Balance intensity and credit exposure
Q04	Pays full balance	Transactor versus balance-carrying behavior
Q05	Card share of purchases	Engagement and share of wallet
Q10	BNPL awareness and use	Adoption of newer digital-credit products

Note. Demographic and financial variables were excluded from class formation and used only for external profiling and validation.

Table 3*Latent Class Model Comparison*

Classes	Log likelihood	AIC	BIC	Decision
1	-17,164.60	34,379.19	34,520.37	
2	-16,964.66	34,031.33	34,319.31	
3	-16,767.80	33,689.61	34,124.41	
4	-16,625.06	33,456.11	34,037.74	Selected
5	-16,571.29	33,400.58	34,129.02	
6	-16,536.70	33,383.41	34,258.66	
7	-16,504.06	33,370.11	34,392.19	
8	-16,473.64	33,361.29	34,530.18	
9	-16,448.67	33,363.34	34,679.05	
10	-16,426.55	33,371.10	34,833.63	

Note. BIC is the primary selection criterion; lower values indicate a better balance between fit and model complexity.

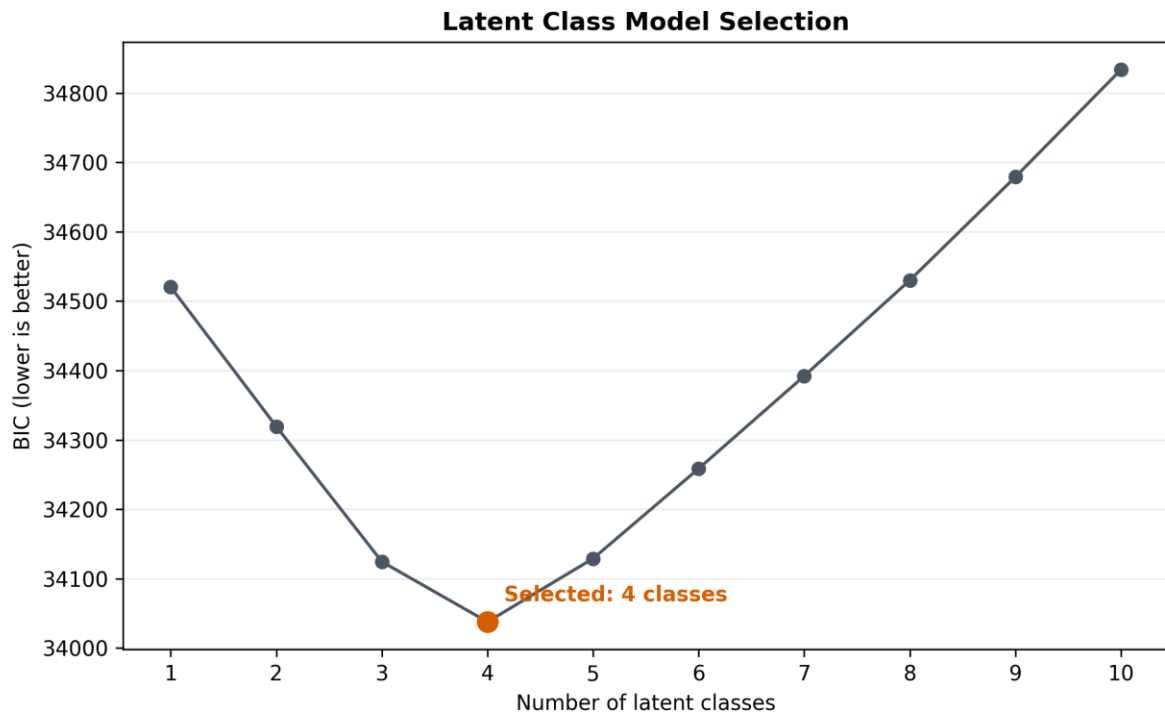
Table 4*Segment Size, External Profile, and Recommended Treatment*

Segment and size	Measured profile	Recommended treatment
High-Use Transactors 637 (30.4%)	Always pay in full: 63.4%; at least 80% card share: 58.8%; high credit score: 71.7%.	Premium rewards, protection, personalized merchant offers, and retention recognition.
Light or Occasional Users 667 (31.9%)	One card: 41.4%; balance below \$500: 52.7%; less than 20% card share: 45.4%.	Low-pressure activation, simple category rewards, wallet onboarding, and fee transparency.
BNPL-Enabled Growth Users 334 (16.0%)	BNPL a few/many times: 96.6%; usage increased: 71.0%; under age 35: 41.3%.	Digital installment offers with total-cost disclosure, reminders, and budget controls.
Balance-Carrying Revolvers 456 (21.8%)	Rarely/never pay in full: 70.7%; balance above \$10,000: 25.6%; financial strain: 28.1%.	Balance alerts, lower-cost repayment options, hardship support, and restrained cross-selling.

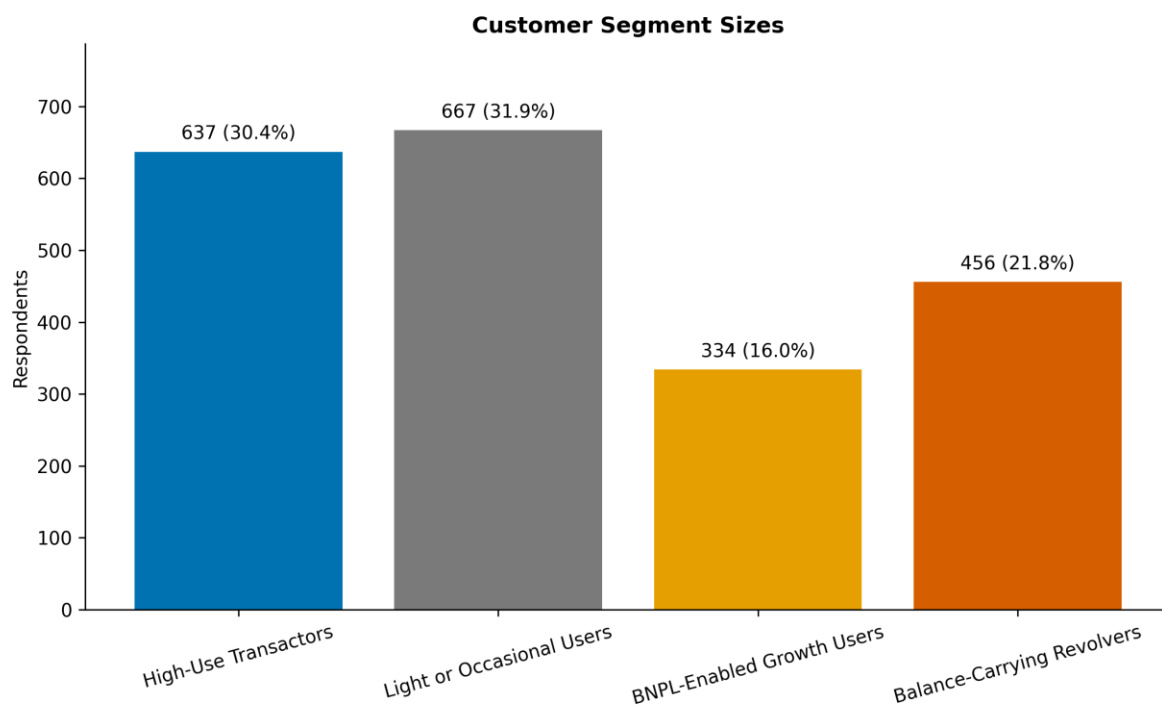
Note. Percentages describe the eligible analytical sample. Marketing treatments should be tested experimentally and should not be used for adverse credit decisions.

Figure 1

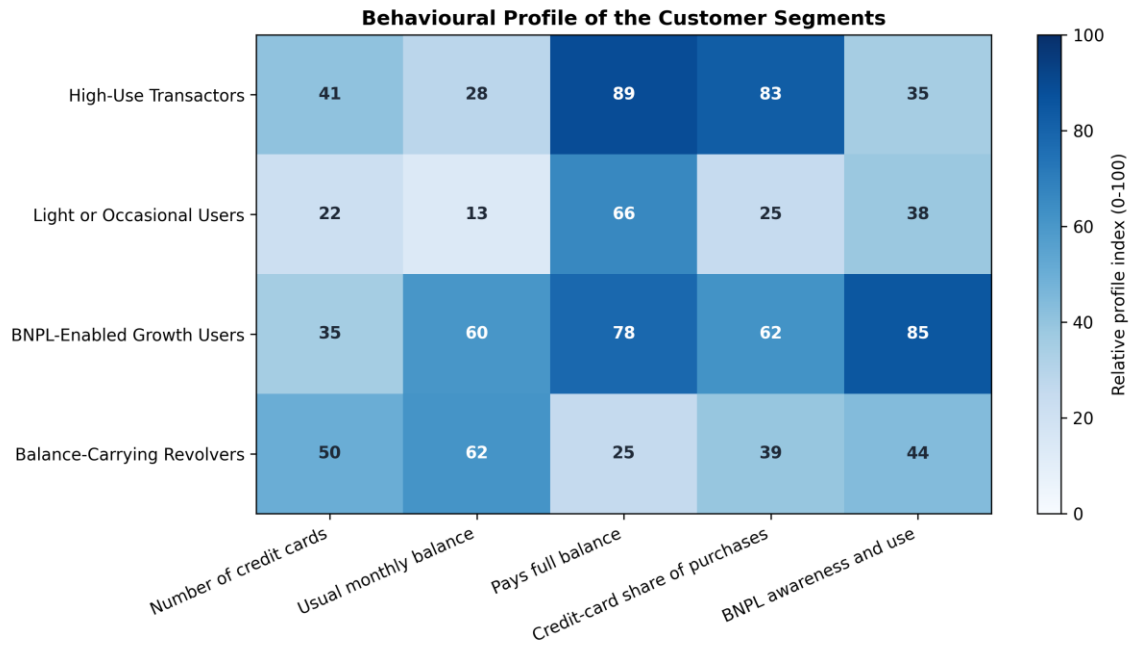
BIC Evidence for the Number of Latent Classes



Note. The minimum BIC occurs at four classes.

Figure 2*Assigned Customer Segment Sizes*

Note. Bars show modal posterior assignments among 2,094 eligible respondents.

Figure 3*Behavioral Profile of the Four Customer Segments*

Note. Indices summarize expected position within each response scale; they are comparative indices, not monetary values.

Appendix A

Proposed Enrichment Variable Framework

Domain	Example variables	Segmentation value
Current behavior	Recent payment mix, card share, merchant categories, recency, frequency, monetary value	Behavioral and value segmentation
Digital engagement	App sessions, digital wallet, email/app response, preferred service channel	Engagement and channel targeting
Psychographics	Convenience, price sensitivity, risk tolerance, debt attitude, financial confidence	Needs-based and attitudinal segmentation
Relationship	Satisfaction, loyalty, advocacy, switching intention, product gaps	Retention and cross-sell prioritization
Life and financial context	Current income, employment change, household stage, goals, emergency resilience	Life-stage relevance and vulnerability safeguards
Governance	Consent scope, collection date, source, version, retention and deletion status	Privacy-compliant use and auditability

Note. The exact survey instrument should be finalized only after cognitive interviews and pilot testing.

Appendix B

Analytical Reproducibility and Decision Rules

Analytical population. The model includes 2,094 active credit-card users with complete Q02, Q03, Q04, Q05, and Q10 responses. Structural routing blanks were not imputed.

Model search. LCA models with one through 10 classes were compared using BIC. Each model used 50 randomized starts, and the minimum-BIC solution was selected.

Classification. Respondents were assigned to the class with the highest posterior probability. Assignments below .60 were retained but flagged for review.

Validation. Demographic and financial variables were excluded from class formation, then assessed through profile percentages, chi-square tests, and Cramer's V. Local-independence residuals were adjusted using Holm's method.

Sensitivity. K-means using standardized ordered category codes was retained only as a robustness comparison because Euclidean distance is not the preferred measurement model for these categorical indicators.

Operational use. Segment labels are descriptive business names applied after estimation. New campaigns should be evaluated using randomized tests, segment drift should be monitored, and uncertain assignments should receive broad rather than highly personalized treatment.